

Oracle Core DBA

Interview Study Guide

Architecture • Storage • Backup & Recovery • Performance
High Availability • Security • RAC • Networking
Patches & Upgrades • Space Management • SQL/PL/SQL Tuning
Multitenant CDB/PDB • Scheduler • AWR / ADDM / ASH

Internals + Interview Q&A; — All Levels

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1. Oracle Architecture

Oracle separates the **instance** (memory + processes) from the **database** (files on disk). Understanding this distinction is the foundation of every DBA topic.

1.1 System Global Area (SGA)

Shared memory allocated when the instance starts. Shared by all server and background processes.

Component	Purpose	Key Parameter
DB Buffer Cache	Caches data blocks read from disk (LRU managed). Sub-caches: DEFAULT_POOL_SIZE, RECYCLE.	DB_BUFFER_CACHE_SIZE
Shared Pool	Library cache (parsed SQL + plans) + Dictionary cache (metadata). Skips redo log writes.	SHARED_POOL_SIZE
Redo Log Buffer	Circular staging area. LGWR flushes on commit, 1/3 full, 3-sec timeout, LOG_BUFFER write.	LOG_BUFFER
Large Pool	RMAN channels, parallel query, shared server (MTS). Separate from shared pool.	LARGE_POOL_SIZE
Java Pool	Java stored procedures and JDBC in-server.	JAVA_POOL_SIZE
Streams Pool	Oracle Streams, GoldenGate capture/apply processes.	STREAMS_POOL_SIZE

Check SGA: `SELECT component, current_size/1024/1024 mb FROM v$sga_dynamic_components ORDER BY current_size DESC;`

1.2 Program Global Area (PGA)

- Private memory for each server process — never shared.
- Contains: sort area, hash area, bitmap merge area, session state, bind variable values.
- Sort/hash exceeding PGA work area spills to TEMP tablespace (one-pass or multi-pass).
- Tune with PGA_AGGREGATE_TARGET (ASMM) or PGA_AGGREGATE_LIMIT (hard cap, 12c+).

1.3 Critical Background Processes

Process	Responsibility	Key fact
DBWR (DBW0..9)	Writes dirty buffers from DB buffer cache to datafiles.	File writer — triggers on checkpoint, buffer pressure, or timeout. Up to 36 writers.
LGWR	Writes redo log buffer to online redo log files.	Synchronous with COMMIT. Defines transaction durability. Put redo logs on fastest disk.
CKPT	Signals DBWR + updates datafile headers and control file with checkpoint.	Does NOT write data itself. Controls recovery time.
SMON	Instance recovery at startup; coalesces free extents; cleans up (applies redo) on roll back (undo uncommitted). DB opens after roll-forward.	Rolls forward (redo) and roll back (undo uncommitted). DB opens after roll-forward.
PMON	Cleans up failed user processes; releases locks; registers every 3 sec with OS.	Dead connection cleanup.
ARCn	Archives filled online redo log groups to archive destinations in ARCHIVELOG mode.	Up to 30 ARCn processes.
MMON	AWR snapshot collection every 60 min; ADDM analysis data in SYS tablespace (DBA_HIST_* tables).	Stays in SYS tablespace.
MMNL	Flushes Active Session History to disk when memory pressure is high.	Enables ASH analysis.

■ **INTERVIEW TRAP: Commit does NOT write data blocks to disk. LGWR writes redo (durability). DBWR writes data blocks asynchronously later. This is the #1 architecture interview trap.**

Architecture Interview Q&A;

Q: Instance crashes mid-transaction. Walk me through recovery.

A: SMON fires on startup → reads control file for last checkpoint SCN → replays all redo from that SCN forward (roll forward) → opens the database for use → rolls back all transactions with no matching COMMIT in redo (roll back) using undo segments. The DB opens as soon as roll-forward completes — roll-back happens after open, in the background.

Q: What is the difference between a latch and a lock (enqueue)?

A: Latch: lightweight serialization for SGA structures. Not queued — spin-and-sleep. Very short duration (microseconds). Examples: library cache latch, cache buffers chains latch. Lock/Enqueue: queued serialization for user-visible resources (rows, tables, transactions). Fair — requesters wait in order. Duration spans the transaction. Examples: TX (row lock), TM (table DML lock). High latch contention = SGA memory pressure. High lock contention = application concurrency problem.

2. Storage — Hierarchy, Segments, UNDO, HWM

Hierarchy (largest → smallest): Tablespace → Segment → Extent → Block

- **Block** — smallest I/O unit. DB_BLOCK_SIZE bytes (default 8KB). Set at creation — permanent. Contains: block header (ITL slots), free space (PCTFREE), row data.
- **Extent** — contiguous set of blocks. Oracle allocates a new extent when a segment needs more space.
- **Segment** — one object's complete storage (table, index, undo, temp). Spans one or more extents.
- **Tablespace** — named logical container for segments. Backed by one or more datafiles.

2.1 Block Internals

- **ITL (Interested Transaction List):** each concurrent DML transaction on the block needs one slot. Default INITRANS=1 for tables, 2 for indexes. Exhausted ITL → enq: TX - allocate ITL entry wait.
- **PCTFREE:** percentage of block reserved for row updates. Default 10%. When a block fills to (100-PCTFREE)%, it's removed from the free list.
- **Row chaining:** row too large for one block → spans multiple blocks. Unavoidable for very wide rows.
- **Row migration:** UPDATE makes row too big → moved to new block; old location has a forwarding pointer → extra I/O on every access. Fix: MOVE the table.

2.2 UNDO Tablespace

- **Read consistency:** queries see data as of their start SCN. Oracle reconstructs old versions from undo blocks.
- **Transaction rollback:** undo restores rows to their pre-DML state.
- **Flashback:** Flashback Query/Table use undo to navigate back in time.
- **Key parameter:** UNDO_RETENTION (default 900s). Increase for long-running queries or flashback.
- **RETENTION GUARANTEE:** set on undo tablespace to prevent Oracle from overwriting retained undo even under space pressure.

■ **INTERVIEW TRAP: ORA-01555 "snapshot too old": undo was overwritten before a long-running query finished. Fix: increase UNDO_RETENTION, increase undo tablespace size, or run long queries during low-activity windows.**

Diagnose: `SELECT maxquerylen, ssolderrcnt FROM v$undostat ORDER BY begin_time DESC;` — ssolderrcnt shows ORA-01555 count per period.

2.3 High Water Mark (HWM)

- **HWM** = highest block ever allocated to a segment. Full table scan reads every block up to HWM — even empty ones.
- **DELETE** never lowers HWM. **TRUNCATE** does. **DROP** does.
- **ASSM** has Low HWM (all blocks formatted) and High HWM (blocks between may or may not be formatted). FTS reads to High HWM.
- `ALTER TABLE t MOVE;` — rebuilds segment, resets HWM. Requires index rebuild. Offline during move.
- `ALTER TABLE t SHRINK SPACE CASCADE;` — online compaction + lowers HWM. Requires `ENABLE ROW MOVEMENT`. ASSM tablespaces only.

Storage Interview Q&A;

Q: You deleted 90% of rows from a large table but full table scans are still slow. Why and how do you fix it?

A: DELETE does not lower the High Water Mark (HWM). Full table scans read every block up to the HWM — even if those blocks are now empty. Fix options: (1) ALTER TABLE t MOVE — rebuilds the segment and resets HWM, but requires downtime (exclusive lock) and index rebuilds after. (2) ALTER TABLE t SHRINK SPACE CASCADE — online compaction, lowers HWM, automatically maintains indexes. Requires ENABLE ROW MOVEMENT and ASSM tablespace. (3) TRUNCATE — resets HWM instantly but deletes all rows. (4) DBMS_REDEFINITION — zero downtime rebuild for very large tables.

3. Backup & Recovery — RMAN, ARCHIVELOG, PITR

3.1 ARCHIVELOG Mode

- Oracle reuses online redo log files in a circular fashion.
- **NOARCHIVELOG**: old redo is overwritten. Can only restore to last full backup — all changes since then are lost.
- **ARCHIVELOG**: ARCN archives each filled redo log before reuse. Full change history preserved.
- Enables: Point-in-time recovery (PITR), hot (online) backups, Data Guard, Streams/GoldenGate.

```
SHUTDOWN IMMEDIATE;  
STARTUP MOUNT;  
ALTER DATABASE ARCHIVELOG;  
ALTER DATABASE OPEN;  
-- Check: SELECT log_mode FROM v$database;
```

3.2 RMAN Backup Types

Type	Description	Use case
Backup Set (default)	RMAN proprietary format. Skips unused blocks. Supports compression and encryption. Must RESTORE before use.	Standard backup strategy.
Image Copy	Block-for-block replica. Same size as original. OS-readable. SWITCH TO COPY with minimal restore needed.	Fast restore to COPY with minimal restore needed.
Level 0 Incremental	All used blocks. Establishes baseline for incrementals. Different from "full" backup.	Initial incremental strategy.
Level 1 Differential	Blocks changed since last Level 0 OR Level 1.	Daily incrementals.
Level 1 Cumulative	Blocks changed since last Level 0 only.	Simpler restore (fewer pieces) but larger daily backup.

✓ **PRO TIP: Incrementally Updated Backup: take an image copy as the base, then apply daily Level 1 incrementals to roll it forward. The copy is always near-current → minimal restore time (no restore step, just SWITCH).**

3.3 Key Recovery Scenarios

Scenario: Datafile lost, DB is open (ARCHIVELOG mode)

1. Offline the file: ALTER DATABASE DATAFILE '/path/file.dbf' OFFLINE;
 2. RMAN> RESTORE DATAFILE '/path/file.dbf';
 3. RMAN> RECOVER DATAFILE '/path/file.dbf';
 4. ALTER DATABASE DATAFILE '/path/file.dbf' ONLINE;
- No downtime required for non-SYSTEM/UNDO datafiles.

Scenario: Point-in-time recovery (PITR)

RMAN> RESTORE DATABASE UNTIL TIME "TO_DATE('2024-01-15 14:30','YYYY-MM-DD HH24:MI')";
RMAN> RECOVER DATABASE UNTIL TIME ... USING BACKUP CONTROLFILE;
SQL> ALTER DATABASE OPEN RESETLOGS;
RESETLOGS creates a new incarnation — take a full backup immediately after!

3.4 Fast Recovery Area (FRA)

- Oracle-managed disk location for: archived logs, RMAN backups, flashback logs, control file autobackups.
- Set: DB_RECOVERY_FILE_DEST and DB_RECOVERY_FILE_DEST_SIZE.
- When FRA fills: database hangs trying to archive. Monitor: SELECT * FROM v\$recovery_area_usage;
- Retention: CONFIGURE RETENTION POLICY TO RECOVERY WINDOW OF 7 DAYS;

Backup & Recovery Interview Q&A;

Q: What happens after you open the database with RESETLOGS?

A: RESETLOGS creates a new database incarnation with a new resetlogs SCN. Archived logs from the previous incarnation cannot be applied to the new incarnation — they belong to a different history. This means: (1) You must take a full backup immediately after RESETLOGS. (2) Any standby databases must be re-synced. (3) RMAN tracks incarnations — you can list them with LIST INCARNATION; and switch back with RESET DATABASE TO INCARNATION if needed (to re-do a botched recovery).

4. Performance — Execution Plans, Bind Variables, Waits

4.1 Reading Execution Plans

- Always use actuals: `SELECT * FROM TABLE(DBMS_XPLAN.DISPLAY_CURSOR(sql_id, 0, 'ALLSTATS LAST +PEEKED_BINDS'))`;
- Read from most-indented operation outward — innermost executes first.
- E-Rows vs A-Rows gap > 10x = bad stats or data skew — the #1 plan problem.

Operation	Meaning	Good when
TABLE ACCESS FULL	Reads every block up to HWM.	Large % of rows, small tables, parallel scan.
INDEX RANGE SCAN	B-tree traversal for a range of values.	Selective WHERE clause, range predicates.
INDEX UNIQUE SCAN	Single row by unique/PK key.	Equality on unique/PK column.
HASH JOIN	Smaller table builds hash table in PGA; larger probes.	Large-table joins, no index on join column.
NESTED LOOPS	Outer row drives, inner row looked up per outer row.	OLTP, small outer result set, index on inner.
SORT MERGE JOIN	Both sides sorted, then merged.	Pre-sorted inputs, inequality joins.
INDEX FAST FULL SCAN	Reads all index blocks (no table access).	COUNT, cover queries — index has all needed columns.

4.2 Bind Variable Peeking & Adaptive Cursor Sharing

- **Bind peeking**: at first hard parse Oracle sniffs the bind value, uses it with histograms to estimate cardinality, builds a plan. That plan is reused for ALL subsequent binds — even ones with very different selectivity.
- **Adaptive Cursor Sharing (11g+)**: monitors bind-sensitive cursors. Creates multiple child cursors with different plans for different bind value ranges. `IS_BIND_SENSITIVE` and `IS_BIND_AWARE` in `V$SQL`.
- **SQL Plan Baselines (SPM)**: lock in the known-good plan. New plans go into history as non-accepted until evolved. Use `DBMS_SPM`.

4.3 Key Wait Events

Wait Event	Cause	Fix
log file sync	Commit waiting for LGWR to flush redo to online redo log.	Batch commits, faster redo log I/O (NVMe), fewer commits.
db file sequential read	Single-block I/O (index reads, undo reads).	Check index efficiency, clustering factor, I/O subsystem.
db file scattered read	Multiblock I/O (full table scan, index FFS).	Expected for FTS. High = too many FTS on large tables.
library cache lock	DDL on an object another session is executing.	Find and resolve the DDL. Check <code>DBA_DDL_LOCKS</code> .
library cache pin	Executing object being recompiled by another session.	Resolve the recompiling session. Recompile objects separately.
enq: TX – row lock	Two sessions updating the same row.	Find blocker in <code>V\$SESSION.BLOCKING_SESSION</code> . Kill blocker.
enq: TX – ITL entry	Block has no free ITL slots for a new transaction.	Increase <code>INITRANS</code> on the table/index.
gc buffer busy (RAC)	Block being transferred via Cache Fusion between nodes.	Reduce cross-node access. Partition by node affinity.
row cache lock	Dictionary cache (sequence nextval) contention.	Increase sequence <code>CACHE</code> value significantly.

Performance Interview Q&A;

Q: E-Rows=1 but A-Rows=450,000 in an execution plan. What do you do?

A: The optimizer drastically underestimated rows (450,000x off). It likely chose a plan optimal for 1 row (nested loops + index) that is terrible for 450,000 rows (should be hash join + FTS or parallel). Root causes: stale or missing statistics, no histogram on a skewed column, implicit type conversion. Fix steps: (1) Gather fresh stats: `DBMS_STATS.GATHER_TABLE_STATS` with `METHOD_OPT=>'FOR ALL COLUMNS SIZE AUTO'`. (2) Check if a histogram is needed on the filter column. (3) If stats are correct but plan is still wrong, use a SQL Profile (via SQL Tuning Advisor) or load a SQL Plan Baseline with the good plan. (4) Short-term: add a hint `/*+ LEADING(t) USE_HASH(t) */` to force the right join method while investigating.

5. High Availability — Data Guard & RAC

5.1 Data Guard

- Maintains one or more standby databases synchronized with primary by shipping and applying redo.
- **Physical standby**: block-for-block copy via Redo Apply (MRP process). Active Data Guard = open read-only while applying.
- **Logical standby**: SQL Apply converts redo to SQL. Different physical structure allowed. Not all data types supported.

Protection Mode	Transport	Primary halts if standby lost?	Data Loss
Maximum Performance	ASYNC	No	Possible (last shipped redo)
Maximum Availability	SYNC (falls back to ASYNC)	No (fallback)	Minimal
Maximum Protection	SYNC AFFIRM	YES — primary shuts down	Zero

■ **INTERVIEW TRAP: Switchover (planned, no data loss, both DB participate) vs Failover (unplanned, primary unavailable, data loss depends on mode). Failover requires old primary to be reinstated as standby.**

5.2 RAC (Real Application Clusters)

- Multiple instances share one database on shared storage (ASM). Each instance has its own SGA and background processes.
- **Cache Fusion**: blocks transferred directly between instance buffer caches over the private interconnect via GCS (Global Cache Service).
- **GES** (Global Enqueue Service): coordinates distributed locks across all instances.
- Voting disk: node eviction decisions — majority wins. OCR (Oracle Cluster Registry): cluster configuration storage.
- **Hot block problem**: sequence with small CACHE causes all instances to contend on the same index leaf block. Fix: increase CACHE, use NOORDER, or use application UUIDs.

```

srvctl status database -d orcl
srvctl add service -d orcl -s oltp_svc -r orcl1 -a orcl2
-- Global V$ views: gv$session, gv$sql (INST_ID column identifies node)
SELECT inst_id, event, count(*) FROM gv$session WHERE wait_class!='Idle' GROUP BY inst_id,event;

```

HA Interview Q&A;

Q: What happens if the RAC interconnect goes down?

A: A node unable to communicate with its peers via the interconnect is evicted from the cluster by CSS (Cluster Synchronization Services). This is called node fencing or node eviction — it prevents split-brain (two nodes both believing they are the active primary). The surviving majority re-forms the cluster and performs instance recovery for the evicted node's in-flight transactions. The evicted node reboots and re-joins. Interconnect health is monitored via CSS heartbeats — if a node misses the CSS timeout (typically 30 seconds of missed heartbeats), it is evicted.

6. Security & Diagnostics

6.1 Privilege Model

Concept	Detail
System privilege	Action on any object of a type. E.g. CREATE TABLE, ALTER ANY TABLE. Granted with ADMIN OPTION (cascading allow)
Object privilege	Action on a specific object. E.g. SELECT ON hr.employees. Granted with GRANT OPTION (cascading allow)
Role	Named collection of privileges. Can be enabled/disabled per session. NOT active inside definer's-rights PL/SQL.
AUTHID DEFINER (default)	Procedure runs with owner's direct privileges. Roles NOT active. Most PL/SQL.
AUTHID CURRENT_USER	Procedure runs with caller's privileges including roles. Used for schema-independent utilities.
WITH ADMIN OPTION	System privilege: grantee can regrant. Revoking from A does NOT revoke from B (A granted to B).
WITH GRANT OPTION	Object privilege: grantee can regrant. Revoking from A DOES revoke from B (cascade).

6.2 Diagnostic Queries

```
-- Find blocking sessions
SELECT sid, blocking_session, event, seconds_in_wait FROM v$session WHERE blocking_session IS
NOT NULL;

-- Kill a session
ALTER SYSTEM KILL SESSION 'sid,serial#' IMMEDIATE;

-- Top wait events right now
SELECT event, count(*) FROM v$session WHERE wait_class!='Idle' GROUP BY event ORDER BY 2 DESC;

-- Invalid objects (breaks library cache)
SELECT owner, object_name, object_type FROM dba_objects WHERE status='INVALID';

-- Recompile all invalid objects
EXEC UTL_RECOMP.RECOMP_PARALLEL(4);
```

7. Oracle Networking — Listener, TNS, JDBC, DRCP

Connection flow: Client → TNS (tnsnames.ora) → Listener → Server process

Component	Purpose	Location
tnsnames.ora	Maps alias → host:port:service. Client-side.	\$ORACLE_HOME/network/admin/ or \$TNS_ADMIN
listener.ora	Listener config: endpoints, static registrations.	\$ORACLE_HOME/network/admin/ on server
sqlnet.ora	Client/server config: encryption, timeouts, auth.	\$ORACLE_HOME/network/admin/ on both sides
Listener process	Accepts incoming connections, hands off to server process. Not in data dictionary.	Server OS process (lsnrctl start)
Dynamic registration	PMON registers services with listener automatically on startup. Preferred.	Automatic — no listener.ora entry needed
Static registration	Defined in listener.ora SID_LIST. Required for NOMOUNT/MOUNT state (RMAN restore).	\$ORACLE_HOME/network/admin/ listener.ora SID_LISTENER section

Key listener commands:

```
lsnrctl status -- status + registered services
lsnrctl reload -- reload listener.ora without restart
lsnrctl trace admin -- enable tracing
ALTER SYSTEM REGISTER; -- force PMON to re-register (fixes ORA-12514)
```

Connection methods: **tnsnames.ora** (file-based alias), **EZConnect** user/pass@host:port/service (no config file), **EZConnect Plus** (19c+: adds timeouts/wallet inline), **LDAP/OID** (central enterprise directory).

DRCP (Database Resident Connection Pooling): pool of server processes inside the DB. Ideal for stateless web apps (PHP/Python/Node) where each request opens/closes a connection. Enable: EXEC

DBMS_CONNECTION_POOL.START_POOL(); Connect with :POOLED suffix.

SQLNET.EXPIRE_TIME: dead connection detection. Oracle sends a probe packet to the client at this interval (minutes). If client is gone, Oracle cleans up the server process and releases all locks. Set to 10. Critical when firewalls silently drop idle connections.

Networking Interview Q&A;

Q: ORA-12514: TNS listener does not currently know of service. Steps to fix?

A: 1. Check what services are registered: lsnrctl services. 2. Verify SERVICE_NAME in tnsnames matches the DB: SELECT value FROM v\$parameter WHERE name='service_names'; 3. Force re-registration: ALTER SYSTEM REGISTER; 4. If DB is in MOUNT state (RMAN restore), use static registration in listener.ora — PMON cannot register a mounted-only instance. 5. If service_names parameter is wrong: ALTER SYSTEM SET service_names='orcl.example.com';

8. Patches & Upgrades — OPatch, Datapatch, AutoUpgrade

8.1 Patch Types

Type	Description
One-off patch	Fixes a single bug. Applied with OPatch. Can be rolled back.
Release Update (RU)	Quarterly cumulative patch (replaces PSU from 18c). E.g. 19.21.0. Contains all fixes since base release.
Release Update Revision (RUR)	Interim fixes between RUs. Cumulative on the RU.
GI RU	Grid Infrastructure patch (clusterware + ASM). Apply before DB home patch.
OJVM patch	Patches Oracle JVM. Must match DB RU version. Applied via datapatch after OPatch.
Datapatch	SQL-based post-install step. Updates dictionary objects after OPatch installs binaries. Mandatory after every p

8.2 OPatch Key Commands

```
opatch lsinventory -- list installed patches
opatch lsinventory -detail -- verbose with file list
opatch prereq CheckConflictAgainstOHWithDetail -ph /patch/dir
opatch apply /patch/35643107 -- apply one-off patch
opatch rollback -id 35643107 -- rollback one-off
opatch napply /patches_dir -- apply multiple patches

-- After OPatch (always run datapatch):
cd $ORACLE_HOME/OPatch && ./datapatch -verbose

-- Check patch status in dictionary:
SELECT patch_id, action, status FROM dba_registry_sqlpatch ORDER BY action_time DESC;
```

8.3 Patching Approaches

- **In-place:** patch directly on the running Oracle Home. DB must be shut down during binary patching. Simple but risky — rollback is harder.
- **Out-of-place:** create a new Oracle Home, install + patch it, then switch databases. Zero downtime possible with RAC rolling. Recommended for production.
- **Rolling patch (RAC):** patch one node at a time while others serve traffic. Requires out-of-place. Not all patches support rolling — check the patch README.
- **AutoUpgrade:** Oracle's tool for full version upgrades. Handles pre-checks, fixups, upgrade scripts, timezone updates: `java -jar autoupgrade.jar -config cfg -mode upgrade`.
- **Patching order:** always patch GI home first, then DB home. For RAC: all nodes GI first, then all nodes DB home.

Patches Interview Q&A;

Q: What is datapatch and why must it be run after every OPatch operation?

A: OPatch only replaces binary files (Oracle executables, shared libraries). The Oracle data dictionary (SYS schema objects, internal tables, packages) is stored in the database and must be updated separately to match the new binaries. Datapatch (\$ORACLE_HOME/OPatch/datapatch) applies the SQL scripts that come with the patch to update the dictionary. Forgetting datapatch leaves the database in an inconsistent state — binaries and dictionary are out of sync. Check status: `SELECT patch_id, action, status, description FROM dba_registry_sqlpatch ORDER BY action_time DESC`. If datapatch was skipped: run it while the DB is open in normal mode.

9. Space Management — HWM, Shrink, Compression, Indexes

9.1 HWM — Reclaim Methods Compared

Method	Online?	Resets HWM?	Index rebuild?	Notes
ALTER TABLE t MOVE	No (exclusive lock)	Yes	Yes (ROWIDs change)	Fastest, simplest. Requires downtime window.
ALTER TABLE t SHRINK SPACE CASCADE	Yes	Yes	No (maintained)	ASSM only. Requires ENABLE ROW MOVEMENT. Online DML allowed.
DBMS_REDEFINITION	Yes (zero downtime)	Yes	Yes (rebuilds)	Complex. Uses MV sync. Best for 24/7 critical tables.
TRUNCATE	Yes	Yes	No	Deletes all rows. Only for purge scenarios.

9.2 Compression Types

Type	Compresses	Licence	Best for
Basic Compression	Direct path inserts only. DML after decompression.	Enterprise	Data warehouses: load-once, read-many.
OLTP (Advanced) Compression	All DML (inserts, updates).	Advanced Compression	OLTP tables with moderate update rate.
Hybrid Columnar Compression (HCC)	By column group within compression units.	Exadata Cloud only	Immutable historical data. Archive tables.

9.3 Index Maintenance

- **Myth busted:** Oracle B-tree indexes self-balance. Routine rebuilding is NOT needed and often counterproductive.
- Rebuild IS needed after: `ALTER TABLE MOVE` (ROWIDs change), tablespace migration, very high deleted-entry ratio (>20-30%).
- Check delete ratio: `ANALYZE INDEX idx VALIDATE STRUCTURE;` then `SELECT del_lf_rows/lf_rows*100 pct_deleted FROM index_stats;`
- `ALTER INDEX idx REBUILD ONLINE;` — rebuilds while allowing DML (Enterprise Edition).
- `ALTER INDEX idx COALESCE;` — merges leaf blocks without lowering HWM. Cheaper than rebuild, faster.

10. SQL & PL/SQL Tuning

10.1 SQL Hints Reference

Hint	Effect
/*+ FULL(t) */	Force full table scan on table t.
/*+ INDEX(t idx_name) */	Force use of a specific index.
/*+ INDEX_FFS(t idx) */	Fast full index scan (reads all index blocks, no table access).
/*+ NO_INDEX(t) */	Prevent any index use on table t.
/*+ PARALLEL(t, 4) */	Use 4 parallel query slaves for table t.
/*+ USE_HASH(t1 t2) */	Force hash join between t1 and t2.
/*+ USE_NL(t1 t2) */	Force nested loops join.
/*+ LEADING(t1 t2) */	Force join order: t1 is the outer/driving table.
/*+ FIRST_ROWS(10) */	Optimize for returning the first 10 rows quickly (OLTP).
/*+ ALL_ROWS */	Optimize for total throughput (default since 10g).
/*+ GATHER_PLAN_STATISTICS */	Collect actual row counts in the plan — use with DISPLAY_CURSOR ALLSTATS.
/*+ CARDINALITY(t 100) */	Override cardinality estimate for table t.
/*+ NO_MERGE(v) */	Prevent view/subquery merging — keep as separate step.
/*+ OPT_PARAM('_optim_peek_user_binds' 'false') */	Disable bind peeking for this statement only.

■ **INTERVIEW TRAP:** Hints are silently ignored if they reference wrong aliases, non-existent indexes, or are physically impossible. Always verify with DBMS_XPLAN — look for "Note: hint not used" in the plan output.

10.2 SQL Plan Management (SPM) vs SQL Profile

	SQL Profile	SQL Plan Baseline
What it stores	Additional cardinality statistics to correct optimizer estimates.	One or more accepted execution plans.
Optimizer freedom	Optimizer still chooses the plan — just with better info. Plans can change.	Only plans in the accepted baseline are used. New plans are not accepted.
Created by	SQL Tuning Advisor (DBMS_SQLTUNE).	DBMS_SPM.LOAD_PLANS_FROM_CURSOR_CACHE
Plan evolution	Not applicable.	New plans must be evolved (verified faster) before becoming accepted.
Use when	Optimizer has wrong estimates but you trust it to choose the right plan.	You need plan stability and want to control exactly which plan is used.

10.3 PL/SQL Performance — BULK COLLECT + FORALL

```
-- SLOW: row-by-row (N context switches between PL/SQL and SQL engine)
FOR r IN (SELECT * FROM orders WHERE status='P') LOOP
  UPDATE orders SET status='D' WHERE id=r.id;
END LOOP;

-- FAST: bulk (1 context switch)
DECLARE
  TYPE id_tab IS TABLE OF orders.id%TYPE;
  l_ids id_tab;
BEGIN
  SELECT id BULK COLLECT INTO l_ids FROM orders WHERE status='P';
  FORALL i IN 1..l_ids.COUNT
    UPDATE orders SET status='D' WHERE id=l_ids(i);
```

```
END;
```

11. Multitenant — CDB, PDB, Local Undo, Clone

11.1 Architecture

- **CDB** (Container Database): the Oracle instance. One set of background processes + SGA shared across all PDBs.
- **CDB\$ROOT**: the root container. Holds common users (C## prefix), common roles, system dictionary.
- **PDB\$SEED**: read-only template. New PDBs cloned from this. Never open for writes.
- **PDB** (Pluggable Database): isolated application database. Own users, schemas, local tablespaces. Up to 4096 PDBs per CDB (19c).
- **Common user**: exists in root and all PDBs. Must be prefixed C##. Privileges granted commonly or locally.
- **Local user**: exists only in one specific PDB. Regular schemas (HR, SCOTT) are local.

11.2 Key Commands

```
-- PDB status
SELECT con_id, name, open_mode FROM v$pdb;

-- Open/close/save state
ALTER PLUGGABLE DATABASE pdb1 OPEN;
ALTER PLUGGABLE DATABASE ALL OPEN;
ALTER PLUGGABLE DATABASE pdb1 SAVE STATE; -- auto-open on CDB restart

-- Create PDB from seed
CREATE PLUGGABLE DATABASE pdb1 ADMIN USER pdbadmin IDENTIFIED BY pw
FILE_NAME_CONVERT=('/pdbseed/', '/pdb1/');

-- Clone PDB
ALTER PLUGGABLE DATABASE pdb1 OPEN READ ONLY;
CREATE PLUGGABLE DATABASE pdb1_clone FROM pdb1
FILE_NAME_CONVERT=('/pdb1/', '/pdb1_clone/');

-- Unplug / plug
ALTER PLUGGABLE DATABASE pdb1 UNPLUG INTO '/tmp/pdb1.xml';
CREATE PLUGGABLE DATABASE pdb1 USING '/tmp/pdb1.xml' NOCOPY;

-- Switch container (from CDB root)
ALTER SESSION SET CONTAINER = pdb1;
```

11.3 Local Undo Mode (12.2+)

- In 12.1 all PDBs shared one undo tablespace — PDB-level flashback and PITR were impossible.
- **Local Undo**: each PDB has its own undo tablespace. Enables PDB-level Flashback, PITR, and clean unplug.
- **Enable**: `ALTER DATABASE LOCAL UNDO ON;` (requires CDB restart).
- **Check**: `SELECT property_value FROM database_properties WHERE property_name='LOCAL_UNDO_ENABLED';`

Multitenant Interview Q&A;

Q: What happens to PDBs when the CDB restarts?

A: By default PDBs come up in MOUNTED state after a CDB restart — they do NOT automatically open. To make them auto-open: ALTER PLUGGABLE DATABASE ALL SAVE STATE; (saves each PDB's current open mode). Alternatively, create a database trigger: CREATE OR REPLACE TRIGGER open_pdb AFTER STARTUP ON DATABASE BEGIN EXECUTE IMMEDIATE 'ALTER PLUGGABLE DATABASE ALL OPEN'; END; SAVE STATE is the recommended 19c+ approach.

12. DBMS_SCHEDULER

12.1 Object Hierarchy

Object	Purpose
Program	Defines WHAT to run: PL/SQL block, stored procedure, or external executable. Reusable.
Schedule	Defines WHEN to run: calendar string or event-based trigger. Reusable.
Job	The execution unit. Combines a program + schedule (or defines inline). Can be simple or part of a chain.
Job class	Groups jobs for resource management — assigns service, resource consumer group, logging level.
Chain	Workflow of steps with conditional logic: if Step A SUCCEEDs run Step B, else run Step C.
Window	Time period during which a resource plan activates (e.g. WEEKNIGHT_WINDOW activates batch plan 10pm-6am).
File watcher	Monitors a filesystem location; fires a job when a specified file arrives.

12.2 Calendar Strings & Commands

```
-- Create a job
DBMS_SCHEDULER.CREATE_JOB(
  job_name => 'NIGHTLY_PURGE',
  job_type => 'STORED_PROCEDURE',
  job_action => 'HR.PURGE_OLD_ORDERS',
  repeat_interval => 'FREQ=DAILY;BYHOUR=2;BYMINUTE=0',
  enabled => TRUE);

-- Calendar examples
-- Every Mon-Fri at 6pm: FREQ=WEEKLY;BYDAY=MON,TUE,WED,THU,FRI;BYHOUR=18
-- Last day of month: FREQ=MONTHLY;BYMONTHDAY=-1;BYHOUR=23
-- Every 15 minutes: FREQ=MINUTELY;INTERVAL=15

-- Monitor
SELECT job_name, state, last_run_duration, next_run_date FROM dba_scheduler_jobs;
SELECT job_name, status, error#, additional_info FROM dba_scheduler_job_run_details
WHERE job_name='NIGHTLY_PURGE' ORDER BY actual_start_date DESC;
```

■ **INTERVIEW TRAP:** Scheduler jobs run in a fresh session — roles are NOT active. All privileges must be directly granted. Jobs also do not inherit NLS settings from your interactive session. Test with:
DBMS_SCHEDULER.RUN_JOB('job_name', use_current_session=>FALSE);

13. AWR / ADDM / ASH — Snapshots, Findings, Queries

13.1 AWR Architecture

- MMON background process takes snapshots every 60 minutes (default). Stores in SYSAUX tablespace (DBA_HIST_* tables).
- Retention: 8 days default. Configure:
`DBMS_WORKLOAD_REPOSITORY.MODIFY_SNAPSHOT_SETTINGS(retention=>30*24*60, interval=>30);`
- AWR Baseline: pin snapshots permanently for before/after comparison:
`DBMS_WORKLOAD_REPOSITORY.CREATE_BASELINE(start_snap_id, end_snap_id, 'pre_upgrade');`
- MMNL flushes Active Session History from V\$ACTIVE_SESSION_HISTORY to DBA_HIST_ACTIVE_SESS_HISTORY when buffer is 2/3 full or every 60 seconds.

13.2 Key DBA_HIST Tables

View	Contains
DBA_HIST_SNAPSHOT	Snapshot ID, begin/end time, instance number.
DBA_HIST_SYSSTAT	System statistics per snapshot: logical reads, parses, transactions/sec.
DBA_HIST_SYSTEM_EVENT	Cumulative wait event times per snapshot. Diff two snapshots to get interval waits.
DBA_HIST_SQLSTAT	Per-SQL metrics per snapshot: elapsed, CPU, IO, buffer gets, executions. Divide by executions for
DBA_HIST_SQL_PLAN	Execution plans captured in AWR. Correlate with SQLSTAT via sql_id.
DBA_HIST_ACTIVE_SESS_HISTORY	1-in-10 sampled ASH data (V\$ASH samples every second; 1 in 10 persisted).
DBA_HIST_OSSTAT	OS-level stats: CPU utilisation, memory, I/O at OS level.
DBA_HIST_SEG_STAT	Segment-level I/O, buffer gets, row lock waits, ITL waits.

13.3 Essential ASH Queries

```
-- What is happening right now (last 5 min)
SELECT event, COUNT(*) samples,
ROUND(COUNT(*)/SUM(COUNT(*)) OVER()*100,1) pct
FROM v$active_session_history
WHERE sample_time > SYSDATE - 5/1440
GROUP BY event ORDER BY 2 DESC;

-- Which SQL is burning CPU right now?
SELECT sql_id, COUNT(*) cpu_samples
FROM v$active_session_history
WHERE session_state='ON CPU' AND sample_time > SYSDATE - 1/24
GROUP BY sql_id ORDER BY 2 DESC;

-- Reconstruct what happened at a specific past time window
SELECT sql_id, event, COUNT(*)*10 est_seconds, session_state
FROM dba_hist_active_sess_history
WHERE sample_time BETWEEN
TO_DATE('2024-01-15 02:00','YYYY-MM-DD HH24:MI') AND
TO_DATE('2024-01-15 02:30','YYYY-MM-DD HH24:MI')
GROUP BY sql_id, event, session_state
ORDER BY 3 DESC FETCH FIRST 15 ROWS ONLY;
```

```
-- Top SQL by elapsed time (last 24 hours from AWR)
SELECT s.sql_id,
ROUND(SUM(s.elapsed_time_delta)/1e6/1000,1) total_ela_s,
ROUND(SUM(s.elapsed_time_delta)/NULLIF(SUM(s.executions_delta),0)/1e6,3) ela_per_exec_s,
SUBSTR(q.sql_text,1,80) sql_text
FROM dba_hist_sqlstat s
JOIN dba_hist_snapshot sn USING (snap_id, dbid, instance_number)
JOIN dba_hist_sqltext q ON q.sql_id=s.sql_id AND q.dbid=s.dbid
WHERE sn.begin_interval_time > SYSDATE - 1
GROUP BY s.sql_id, q.sql_text
ORDER BY 2 DESC FETCH FIRST 10 ROWS ONLY;
```

AWR/ADDM Interview Q&A;

Q: Users report the database was slow between 2–3am last night. How do you investigate?

A: 1. Generate AWR report for that window: @\$ORACLE_HOME/rdbms/admin/awrrpt.sql — select snapshots covering 2-3am. Check Load Profile (DB time/sec) and Top Wait Events. 2. If AWR is too coarse, use ASH: query DBA_HIST_ACTIVE_SESS_HISTORY with sample_time BETWEEN the window. Group by sql_id, event to find the dominant wait and SQL. 3. Check for blocking: WHERE blocking_session IS NOT NULL in DBA_HIST_ACTIVE_SESS_HISTORY. 4. If a batch job was running: check DBA_SCHEDULER_JOB_RUN_DETAILS for jobs active in that window. 5. Check ADDM findings for that snapshot pair — they may already point to the issue.

Q: What is DB time and why does Oracle use it as the primary performance metric?

A: DB time = sum of CPU time + non-idle wait time across ALL sessions. It is a measure of total work the database is doing — not wall-clock time. On a system with 100 sessions, DB time can be 100x wall-clock time. Oracle uses DB time because it normalises across parallelism levels and lets you express impact as a percentage of total work. An ADDM finding that says "SQL X consumed 40% of DB time" means 40% of all database work in that period was for that one statement — far more actionable than raw elapsed time. AWR Load Profile shows DB time/second; baseline this metric for your normal workload.

Quick Reference — Essential V\$ Views

View	Use
V\$DATABASE	DB name, log_mode, open_mode, created.
V\$INSTANCE	Instance name, startup time, status, version.
V\$SGA / V\$SGAINFO	SGA component sizes.
V\$SGA_DYNAMIC_COMPONENTS	Current and max sizes of auto-managed SGA components.
V\$SESSION	Current sessions: username, status, sql_id, event, blocking_session.
V\$SQL	Cached SQL: elapsed, cpu, buffer_gets, executions, sql_text.
V\$SQL_PLAN	Execution plans for cached SQL cursors.
V\$ACTIVE_SESSION_HISTORY	Real-time ASH: 1-second samples of active sessions (last ~30 min).
V\$UNDOSTAT	Undo performance stats: maxquerylen, ssolderrcnt (ORA-01555 count).
V\$TRANSACTION	Active transactions: undo used, undo records, start time.
V\$LOCK	Current locks held and requested.
V\$LATCH	Latch statistics: gets, misses, sleeps.
V\$DATAGUARD_STATS	Data Guard lag, apply rate, transport lag.
V\$MANAGED_STANDBY	MRP and RFS process status on standby.
GV\$SESSION	RAC: same as V\$SESSION but across all instances (INST_ID column).
V\$RECOVERY_FILE_DEST	FRA usage: space_limit, space_used, space_reclaimable.
V\$RMAN_BACKUP_JOB_DETAILS	RMAN job history: status, start/end time, input/output size.
DBA_HIST_SNAPSHOT	AWR snapshot registry: snap_id, begin/end time.
DBA_HIST_ACTIVE_SESS_HISTORY	Historical ASH (1-in-10 persisted from V\$ASH).
DBA_SCHEDULER_JOB_RUN_DETAILS	Job execution history: status, duration, error#.
DBA_REGISTRY_SQLPATCH	Patch history: patch_id, action, status, description.
V\$PDBS	CDB: all PDBs with con_id, name, open_mode.

Top Interview Traps — Memorise These

✗ MYTH: "Commit writes data to disk"

✓ REALITY: COMMIT writes redo (via LGWR) to the online redo log — not data blocks. DBWR writes data blocks asynchronously later. Commit = durability via redo, not via data file writes.

✗ MYTH: "Roles work inside PL/SQL procedures"

✓ REALITY: Roles are NOT active inside definer's-rights (AUTHID DEFINER) PL/SQL. All required privileges must be directly granted. This is the #1 privilege trap.

✗ MYTH: "Rebuild indexes regularly for performance"

✓ REALITY: Oracle B-tree indexes self-balance. Routine rebuilding is a myth and can cause performance degradation. Rebuild only after ALTER TABLE MOVE, tablespace migration, or very high deleted-entry ratio.

X MYTH: "DELETE frees space and lowers the HWM"

✓ REALITY: DELETE never lowers the High Water Mark. Full table scans still read all blocks up to the old HWM. Use SHRINK, MOVE, or TRUNCATE to reclaim space.

X MYTH: "AWR and ASH require no licence"

✓ REALITY: AWR and ADDM require the Diagnostics Pack licence. ASH requires both Diagnostics Pack and Tuning Pack. Using these tools without the licence is a violation.

X MYTH: "ARCHIVELOG mode slows the database down significantly"

✓ REALITY: ARCHIVELOG mode adds minimal overhead — only the ARCn process writing archived logs asynchronously. The redo generation overhead already exists in NOARCHIVELOG mode. A properly sized LOG_ARCHIVE_DEST is essentially invisible.

X MYTH: "Switchover and Failover are the same thing"

✓ REALITY: Switchover is planned (both DBs participate, zero data loss). Failover is unplanned (primary unavailable, possible data loss depending on protection mode). After failover, the old primary must be reinstated as a standby.

X MYTH: "WITH ADMIN OPTION and WITH GRANT OPTION behave the same way on revoke"

✓ REALITY: ADMIN OPTION (system privilege): revoking from A does NOT revoke from B (no cascade). GRANT OPTION (object privilege): revoking from A DOES revoke from B (cascades). This asymmetry is a classic exam and interview question.

End of Guide

Good luck with your Oracle DBA interviews!